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OPIOID & ALCOHOL USE DISORDER

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

BAC	Blood Alcohol Content
BID	Twice Daily
BP	Blood Pressure
BZD	Benzodiazepine
CDSA	Controlled Drugs and Substances Act
CIWA-AR	The Clinical Institute Withdrawal Assessment Alcohol Scale, Revised
C/I	Contraindications
CNS	Central Nervous System
CrCl	Creatinine Clearance
CYP	Cytochrome P450
D/I	Drug Interaction
DOC	Drug of Choice
eGFR	Estimated Glomerular Filtration Rate
GERD	Gastroesophageal Reflux Disease
HCTZ	Hydrochlorothiazide
HR	Heart Rate
LFT	Liver Function Test
MAOI	Monoamine Oxidase Inhibitor
MD	Medical Doctor
MMT	Methadone Maintenance Treatment
MOA	Mechanism of Action
MSK	Musculoskeletal
N/V/D	Nausea/Vomiting
NOWS	Neonatal Opioid Withdrawal Syndrome
OAT	Opioid Agonist Treatment
OCP	Ontario College of Pharmacists
PAWSS	Pennsylvania Alcohol Withdrawal Syndrome Score
SE	Side Effect
SL	Sublingual
SUD	Substance Use Disorder
UDT	Urine Drug Test

DEFINITIONS

Substance Use Disorder (SUD) is a chronic, relapsing disorder with long-lasting changes in the brain, caused by maladaptive and repeated misuse of substance(s)

Complicated by Psychosocial aspects…

Physical Dependence

- Withdrawal symptoms from drug discontinuation

Tolerance

- Diminished effect of a drug over time (requiring higher doses to achieve the same effect)

Opioid Toxicity

- Overdose or excessive intake

Characteristics of SUD

- Use more than intended due to cravings
- Use despite negative consequences
- Loss of control of use
- Compulsion to use

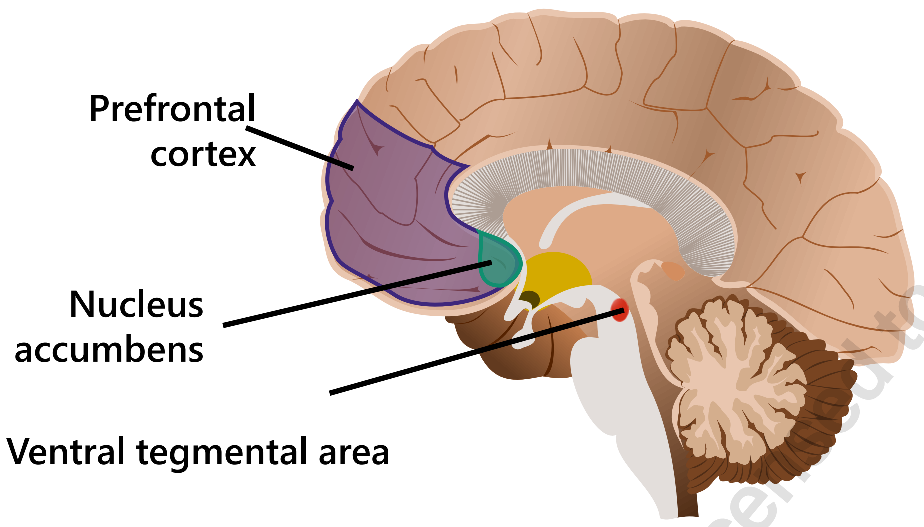
PATHOPHYSIOLOGY

Dopamine

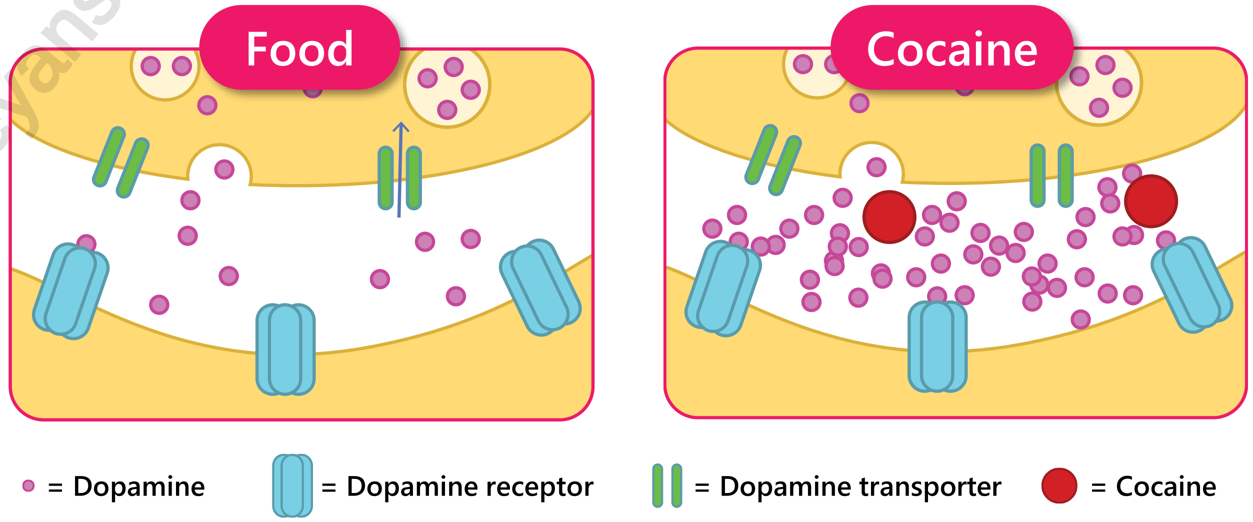
- Regulates movement, emotion, motivation, and feelings of pleasure or euphoric effects
- Reinforces our behavior to repeat pleasurable activities

DRUGS OF ABUSE TARGET THE BRAIN'S PLEASURE CENTER

Brain reward (dopamine) centers



Drugs of abuse increase dopamine



Dopamine increases in response to natural rewards (e.g. food)
When cocaine is taken, dopamine levels are exaggerated

DRUGS WITH ABUSE POTENTIAL

Type	Examples
Depressants:	
1. Opioids	Morphine, codeine, methadone, hydromorphone, oxycodone, buprenorphine, fentanyl, hydrocodone, heroin
2. Benzodiazepines	Clonazepam, lorazepam, alprazolam, temazepam, oxazepam
3. Sleep aids	Zopiclone, zolpidem
4. Barbiturates	Butalbital
5. Other	Alcohol, gabapentinoids, dimenhydrinate
Stimulants	Cocaine, amphetamine, methamphetamine, methylphenidate, nicotine, caffeine
Hallucinogens	LSD (acid), MDMA, ketamine, psilocybin (magic mushrooms), cannabis

HARM REDUCTION

What is harm reduction?

- Harm reduction services are intentional practices that help minimize the negative effects of various actions, harm reduction may include
  - Public health policies
  - New laws or changes to existing laws
  - Community services and programs
- Harm reduction save lives and should always be incorporated into SUD treatment
- Remember, addition or misuse of medications or other substances is not a moral failing of patients

OUD Harm Reduction Examples

- OAT therapy
- Naloxone kits
- Safe consumption sites
- Needle exchange
- Safer supply prescriptions
- Infectious disease testing
- Vaccinations

Other Harm Reduction Examples

- Community resources for food, shelter, etc.
- Peer programs for social support
- Supplying condoms
- Seatbelts in cars
- Drinking water when drinking alcohol

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OPIOID USE DISORDER

Qualifying Exam Preparatory Course

OPIOID USE DISORDER

Overview

- Chronic, relapsing condition characterized by craving, uncontrolled use, and continued use despite significant consequences
- Opioids have a high likelihood of misuse due to their CNS-depressing and analgesic properties, with the potential for euphoric effects
- Associated with increased morbidity and mortality → sustained long-term remission can be achieved with appropriate treatment and care
- Standard of Care: Opioid Agonist Treatment (OAT)
- Patients should be supported to reduce harm through education on safer drug use, take-home naloxone kits, and referral to other harm-reduction services

OPIOID USE DISORDER

NON-PHARMACOLOGICAL MEASURES

- Psychosocial support = crucial aspect of treatment
- Regular contact with healthcare providers
- Interventions that reinforce harm reduction and engagement:
  - Addiction counselling
  - Peer support
  - Contingency management
  - Affordable housing and income support
  - Education: how to safely use sterile syringes, needles, and other substance-use equipment
  - Access to:
    - Sterile syringes, needles, other supplies
    - Take-home naloxone kits
    - Supervised consumption services

TREATMENT OVERVIEW

- Most patients diagnosed with moderate-severe OUD require pharmacotherapy as first-line therapy in addition to psychosocial treatment
- Withdrawal management alone is not recommended, OAT should be offered
- Opioid agonists are selected to be trialled first

- Exact regimen should be individualized
  - Open ended treatment based on patient's goals and circumstances
  - Slow tapering may be considered for patients wishing to stop treatment

Treatment Options	Administration
Buprenorphine/ Naloxone (Suboxone®)	Sublingual tablets or film daily
Buprenorphine ER (Sublocade®)	Subcutaneous injection once monthly
Methadone (methadose®)	Daily drink PO
Morphine SR (Kadian®)	Daily capsules PO
Naloxone	Intranasal or IM, as needed for overdose

OPIOID USE DISORDER

Methadone (Methadose®)

Methadone
- Full opioid agonist
- Some NDMA antagonism

❑ First-line therapy
Alternative agent for opioid use disorder for patients with poor response to buprenorphine/naloxone
No ceiling dose; doses may be split BID in certain circumstances (e.g. pregnancy, rapid metabolizers)

Administration:
- 10 mg methadone/mL stock solution is diluted in a vehicle to make a 100 mL solution
    - Orange drink recommended (e.g. Tang®) to deter injection or diversion (NOT water)
- Slow increase in dosing
- Daily supervised ingestion and confirmation may be required (province/territory-specific)
- If the patient appears intoxicated before administration, defer methadone, document, and inform MD
    - Dosing may need adjustment – safety concern

OPIOID USE DISORDER

Methadone (Methadose®)

Adverse Effects:

- Sedation
- Sweating
- Constipation
- Decreased libido
- Decreased plasma testosterone
- Weight gain
- QTc prolongation
- Nausea
- Insomnia
- Headache

Drug Interactions:

- CYP3A4 inhibitors/inducers
- CNS depressants
- Serotonergic agents
- Anticholinergic agents
- QTc prolonging agents
- MAOI

METHADONE

LEGAL REQUIREMENTS

- ✓ Prescriptions must follow all requirements for “straight narcotics”
  - ✓ Date written
  - ✓ Drug name and dose (milligram dose in words AND numbers)
  - ✓ Start and stop dates stated as inclusive
  - ✓ Dosing regimen (e.g. specify observed AND take-home dose days)
- ✓ Documented log sheet: time and date of dose taken
- ✓ Province/territory-specific requirements may also differ

Prescribers no longer require special exemptions to prescribe

METHADONE

MISSED DOSES

Prescriber must always be notified!

Patients should be supported to resume their usual dose as quickly and safely as possible (this process is individualized and conducted under physician supervision)

Missed 1-2 days	• Give usual dose as long as no concerns of loss of tolerance/adverse events
Missed 3 days	• Patient must be assessed by MD: do not provide dose until seen by MD • May need around a 50% dose decrease due to loss of tolerance
Missed 4 or more days	• Major loss of tolerance • Do not provide dose until seen by MD • May be restarted on low initial doses (generally 30 mg or less)
Emesis (vomiting)	• Vomited doses are not replaced unless witnessed by a healthcare provider • Offer no more than 50% of the dose ONLY if emesis occurs within 15 minutes of ingestion (this may vary based on provincial/territorial standards and guidelines) • Only one replacement dose should be given • Exceptions to the above include special populations: pregnancy, HIV, or cancer • If emesis occurs due to pregnancy, consider spreading the dosing over 30 minutes

OPIOID USE DISORDER

Buprenorphine/Naloxone (Suboxone®)

Suboxone®
- **Buprenorphine:** partial mu opioid agonist and weak kappa antagonist
- **Naloxone:** opioid antagonist activated if crushed/injected → mitigates risk of misuse

❑ First-line therapy
Dosed 2-24 mg/day once daily or alternate day dosing

Administration:
- SL tablet: depending on the province/territory, may need to just watch the patient place it under the tongue or wait until it partially or fully dissolves
- Film: can be placed under the tongue (sublingual) or inside either cheek (buccal) until completely dissolved

Adverse Effects:
- Headache, constipation, sweating, N/V, decreased libido (long-term use), dental effects
  - Patients should take a sip of water, swish it around their teeth and gums, then swallow
  - Wait 1 hour before brushing teeth

Drug Interactions:
- Alcohol, BZD, CNS depressants, opioids, naltrexone, CYP3A4 inhibitors/inducers, MAOI, serotonergic agents

OPIOID USE DISORDER

Buprenorphine/Naloxone (Suboxone®)

Clinical Pearls:

- Ceiling effect of buprenorphine is safer than methadone - less risk of overdose
- Has a lower potential for misuse and a mild withdrawal profile
- May be less sedating and better tolerated than methadone
- Naloxone has poor bioavailability; therefore, it does not cause withdrawal if used as prescribed
- Lower risk of QT prolongation compared to methadone
- Always review province-specific information regarding OAT as it can vary widely (documentation, administration, etc.)

OPIOID USE DISORDER

Buprenorphine Extended Release (Sublocade®)

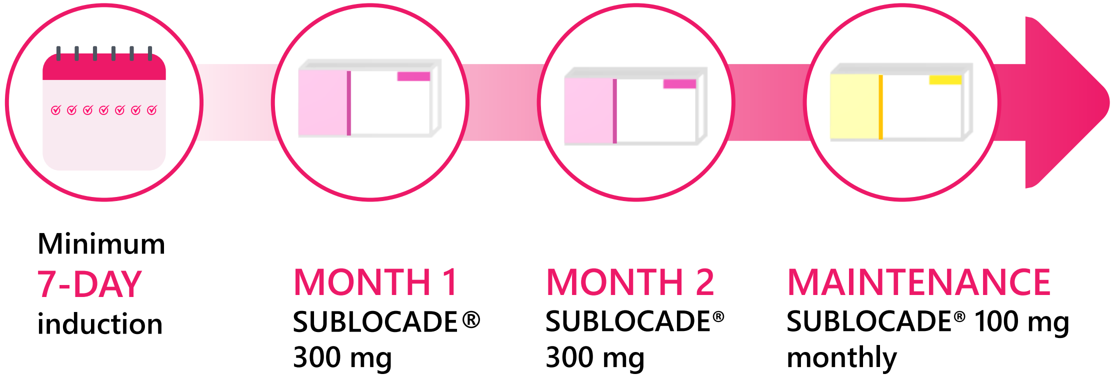
Buprenorphine extended release (subcutaneous injection)	❑ First-line therapy Indicated for the management of moderate-to-severe opioid use disorder in adults

Administration:

- Once-monthly injection that delivers buprenorphine at a continuous, controlled rate
- Patients must be stabilized on Suboxone® (buprenorphine/naloxone) for at least 7 days
- Must be taking between 8 and 24 mg of Suboxone® (buprenorphine/naloxone)
- Starting dose is 300 mg monthly for 2 months
- Maintenance dose is 100mg monthly
    - 26 days minimum between doses

Adverse Effects:

- Nausea, vomiting, constipation, fatigue and injection site pain/pruritus, hypotension



OPIOID USE DISORDER

Slow-Release Oral Morphine (SROM)

Morphine SR
- Full opioid agonist
- Daily dosing (Kadian®)

❑ Second line therapy
When buprenorphine/naloxone or methadone is ineffective or contraindicated
Off-label use for OUD (generally managed by a specialist)

Administration:
- Supervised once-daily witnessed ingestion and confirmation may be required
- If the patient appears intoxicated before administration, defer, document, and inform MD
- Take-home doses may be prescribed if the patient meets certain province-specific criteria

Adverse Effects:
- Headache, abdominal pain, pruritus, dizziness, N/V, constipation and dry mouth

Drug Interactions:
- Pharmacodynamic interactions with concurrent medications that cause respiratory and CNS depression

SUMMARY OF OPIOID USE DISORDER TREATMENT

- Buprenorphine as combination or monotherapy are first-line treatment
- Methadone is first-line treatment
  - Buprenorphine/naloxone is available as a SL tablet and film
  - Buprenorphine is available as monthly injection
  - Methadone is available as a liquid
  - Morphine SR is second line therapy, often prescribed off label

- Always check drug interactions with methadone as it is a substrate of CYP2B6 (major), CYP3A4 (major), CYP2C19 (minor), CYP2C9 (minor), CYP2D6 (minor)

- Always notify MD of any missed doses of these agents
  - If patient has missed a methadone dose for 3 days, do not dispense dose, refer to MD for assessment
  - If vomiting occurs, only replace the dose by 50% if it was witnessed by you AND if it was within 15 min of ingestion

SUMMARY: METHADONE VERSUS BUPRENORPHINE

METHADONE	BUPRENORPHINE/NALOXONE
• No maximum dose	• Less risk of overdose due to ceiling effect for respiratory depression
• Potentially better treatment retention	• Reduced risk of injection, diversion, and overdose due to naloxone component
• May be easier to initiate treatment	• Milder SE profile
• Higher risk of overdose (particularly during initiation)	• Shorter time to achieve therapeutic dose
• More severe side effect profile	• Fewer drug interactions
• Many drug interactions	• Alternate day dosing schedules are possible
• Higher risk of QT-prolongation	• May be suboptimal for individuals with high opioid tolerance
• May need daily pharmacy attendance	• Take home dosing as soon as clinical stability is achieved (generally about 7 to 10 days)

OUD FIRST LINE THERAPY COMPARISON

	Buprenorphine/naloxone (Suboxone®)	Methadone (Methadose®)
Drug Class	Opioid Agonist	
Pearls	• Indicated in patients with moderate-severe OUD with access to a clinician and may require dose witnessing in the pharmacy • Fewer drug interactions • Safer ceiling effect than methadone (lower overdose risk)	• Indicated in patients with higher physical levels of dependence (e.g. fentanyl use) with severe disorders or poor response to buprenorphine/naloxone • More efficacious at retaining patients during treatment • 10mg/mL stock solution to be mixed with orange drink (e.g. tang) to deter injection • 3 missed doses require a provider assessment • Dose replacement of $\leq$50% if emesis is witnessed within 15 minutes
Caution	• Risk of withdrawal when initiating	• Requires slow titration to prevent overdose • QTc prolongation risk
Adverse Effects	• Headache, constipation, dental effects, sweating	• Sedation, decreased libido, weight gain, peripheral edema, insomnia, N/V
Interactions	• Alcohol, BZD, CNS depressants, opioids, MAOI, serotongeric agents • Naltrexone • CYP 3A4 inducers/ inhibitors	• CNS depressants, serotonergic agents, anticholinergic agents, MAOI • CYP 3A4, 2D6, 2B6

OPIOID OVERDOSE

Overdose occurs when a person **uses more of a drug than the body can handle**

Risk Factors

- Mixing drugs or using alone
- Inconsistent drug quality
- Being a new drug user
- Use after period of cessation
- Experienced a previous overdose

Signs of an Overdose

- Slow or shallow breathing
- Deep gurgling or snoring sounds
- Lips, nails are pale or blue
- Pupils are constricted
- Skin is cold and clammy
- Body is limp

Overdose is a medical emergency!
Steps to take if you suspect an overdose:

Stimulation
Are they responsive? Try to wake them up

↓

Call 911 Immediately

↓

Administer Naloxone (4 steps)

Do chest compressions +/- rescue breathing

↓

Recovery Position

NALOXONE

OVERVIEW

Naloxone is an opioid antagonist that can temporarily reverse an opioid overdose

- If used right away, naloxone can help normalize breathing and regain consciousness
- Naloxone can be either injected or given as a nasal spray
- IV naloxone has the fastest onset (1-2 mins)

Eligibility Criterion - Naloxone Kit

- In Ontario, any eligible individual can access nasal spray or injectable naloxone at no cost through a participating pharmacy (no OHIP or ID card required)
- Pharmacists should provide training on administration

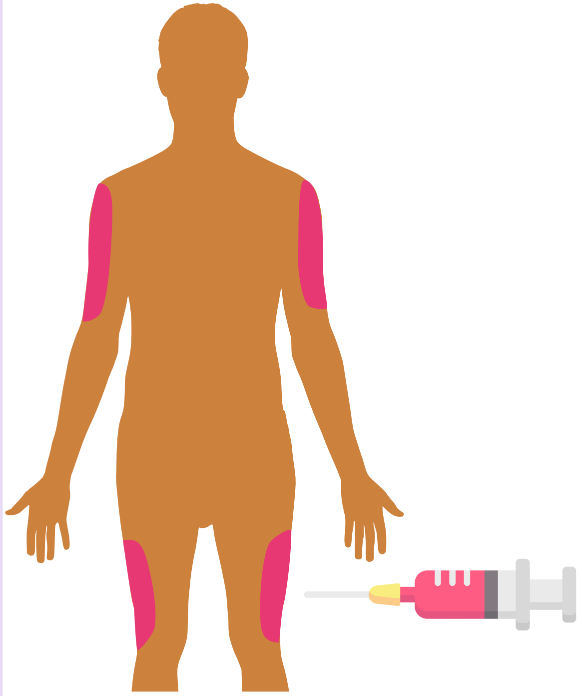
Refills are available from Public Health

NALOXONE

INTRAMUSCULAR ADMINISTRATION (ONSET= 2-5 MINS)

1	Tap the top of the ampoule to release air bubble(s)
2	Break open the ampoule using the snapper (or your sleeve, alcohol swab, etc.)
3	Insert a new syringe into the ampoule and slowly draw up all of the naloxone into the syringe
4	Inject all of the naloxone (1 mL) into upper arm muscle or thigh muscle at a 90-degree angle (IM administration)

Repeat the second dose in 2 to 3 minutes if the first dose does not work



NALOXONE

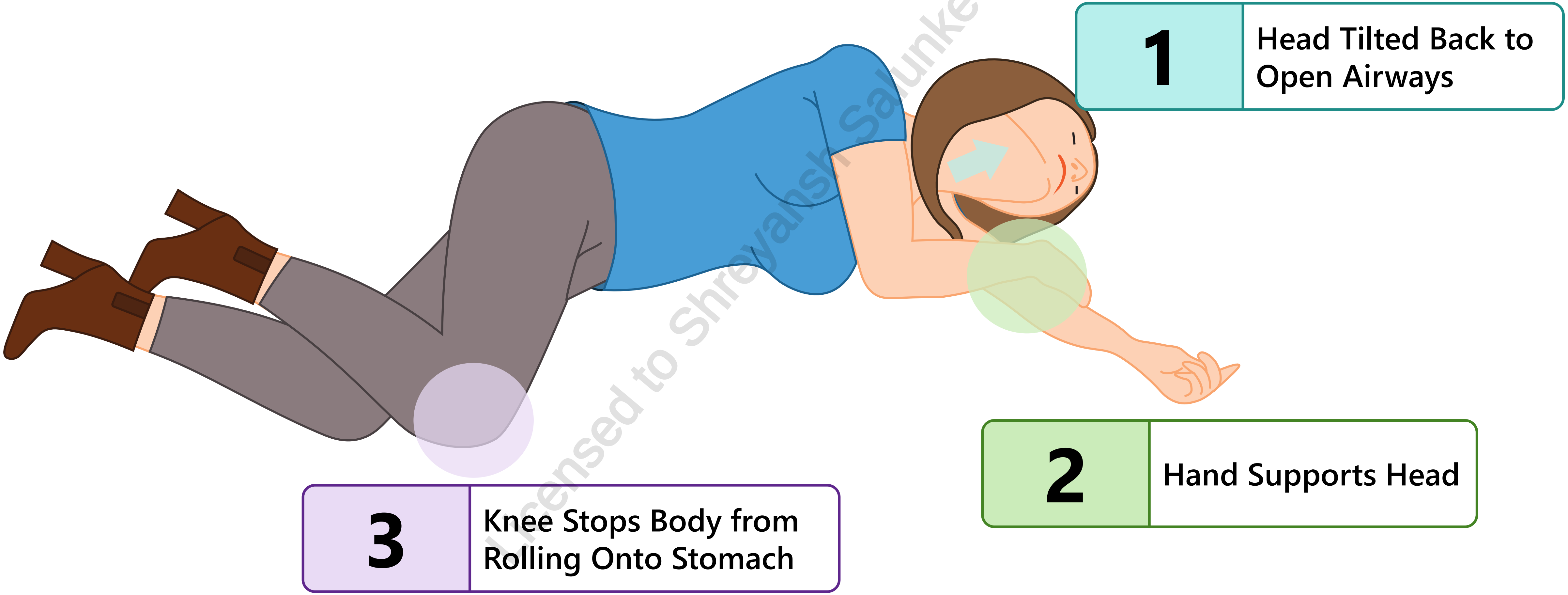
INTRANASAL ADMINISTRATION (ONSET= 2-5 MINS)

1. Lay person on their back, supporting neck and tilting head back
2. Peel the package open and hold the device, do not press until ready to give naloxone
3. Place the tip into the nostril
4. Press firmly

Repeat the second dose in 2 to 3 minutes if the first dose does not work

RECOVERY POSITION

Final Step: place the individual in the recovery position if they are breathing on their own (or if must be left unattended at any time):



SPECIAL POPULATIONS

Therapy in Pregnancy

- Untreated OUD in pregnant patients is associated with increased complications throughout pregnancy and birth
- Benefits of OAT during pregnancy outweigh neonatal risks
- 1st line: methadone, buprenorphine +/- naloxone
  - Methadone can cause neonatal opioid withdrawal syndrome (NOWS) which is treatable
    - Less risk than acute withdrawal or other opioid use
  - Buprenorphine may cause less severe NOWS
  - Unnecessary to transition pregnant patients from their current therapy unless requested or clinical concerns

Therapy in Breastfeeding

- Methadone and buprenorphine/naloxone
- Monitor for sedation and respiratory depression

Therapy in Elderly

- Must be titrated due to risk of sedation and overdose

Switching between methadone and Suboxone® during pregnancy and breastfeeding is not recommended for patients who were stable on one of these medication before becoming pregnant

MONITORING PARAMETERS

EFFICACY ENDPOINTS				
Parameter	Monitoring Tool	Frequency	Degree of Change	Timeframe
Use of other opioids	UDT	As needed (e.g. at OAT initiation, dose adjustment, concerns about patient presentation) → can vary from weekly during stabilization to 1-3 months once stabilized	Minimal to none	Days to weeks
Quality of life	Patient questionnaire	Every visit	Improvement	Weeks to months

MONITORING PARAMETERS

SAFETY ENDPOINTS				
Parameter	**Monitoring Tool**	**Frequency**	**Degree of Change**	**Timeframe**
Missed Doses	Patient questionnaire	Every visit	Minimal to none	Duration of therapy
Sedation and Drowsiness				
N/V				
Respiratory depression	Respiratory rate		None	
QTc prolongation	ECG	Baseline and as needed		

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ALCOHOL USE DISORDER

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ALCOHOL USE DISORDER

OVERVIEW

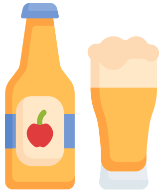
- Defined in the DSM-V as a "problematic pattern of alcohol use leading to clinically significant impairment or distress"
- Goals of therapy: reduce alcohol consumption or achieve abstinence
- Best Practice: Integration of pharmacological and non-pharmacological treatment, including harm reduction strategies
- First-line pharmacological treatment: acamprosate or naltrexone

CANADIAN CENTRE ON SUBSTANCE USE AND ADDICTION RECOMMENDATIONS

- Standard drinks are calculated based on the amount of pure alcohol they contain
- Guidelines to reduce the risk of short- and long-term health effects from alcohol:
  - ❑ Any amount of alcohol can pose health risks
    - No alcohol = No risk
    - ≤ 2 standard drinks per week = low risk
  - ❑ CCSA advises not to exceed 2 drinks on any day for individuals who choose to drink daily, despite the recommendations

Non-pharmacological counselling to help patients stay on target:

- 1:1, for every alcoholic drink, have one non-alcoholic (e.g. water, pop, juice)
- Drink slowly, eat before drinking
- Try to have alcohol- free weeks
- Reduce harm by reducing alcohol use

What is 1 standard drink?	
 Beer 5% alcohol 341 mL (12 oz)	 Coolers or Ciders 5% alcohol 341 mL (12 oz)
 Wine 12% alcohol 142 mL (5 oz)	 Distilled alcohol 40% alcohol 43 mL (1.5 oz)

NON-PHARMACOLOGICAL MEASURES

- Cornerstone in management to be used alone or in combination with pharmacological therapy
- Goals: Behavioural changes, prevent relapse, improve overall function and quality of life

Behavioral therapies

- CBT
- Mindfulness-based intervention

Lifestyle

- Exercise
- Sleep

Support Groups

- Group therapy
- Alcoholic anonymous, 12-step recovery program

Nutritional Support

- Thiamine (vitamin B1)
- B-complex, folate, vitamin C, protein

Motivational enhancement

- Interviewing
- Reinforcing relapse is a normal part of the process

WITHDRAWAL MANAGEMENT

PAWSS is FYI, recommended to know management

Use the PAWSS (Prediction of alcohol withdrawal severity scale)

Low risk of severe complications of alcohol withdrawal

In the outpatient setting, consider:
- Gabapentin
- Carbamazepine
- And/or adjunctive clonidine

Do NOT prescribe:
- Antipsychotics OR SSRI

High risk of severe complications of alcohol withdrawal

Ideally, in the inpatient setting, consider:
- Short-term BZD
- Caution: Do not prescribe BZD as ongoing treatment

GUIDELINE DIRECTED THERAPY

1. Treatment Recommendations: Trial first-line therapy: naltrexone or acamprosate for at least 6 months
2. Titrate or switch drug class as needed
3. Consider Second line therapy

Naltrexone

- Indication: patients with a treatment goal of abstinence OR reduction in alcohol consumption
- Contraindications: Any current opioid use/ opioid withdrawal, acute hepatitis or liver failure
- Caution: Renal impairment, severe haptic impairment or concurrent use of hepatoxic drugs, pregnancy and breastfeeding, patients under the age of 18
- Adverse effects: Nausea, dizziness, headache; symptoms can improve with abstinence
- Dosing (FYI): 25mg once daily for 3 days, then titrate to 50mg once daily over 2 weeks

Acamprosate

- Indication: patients with the treatment goal of abstinence (NOT for reduction only)
- Contraindications: severe renal impairment, breastfeeding
- Caution: moderate renal impairment (dose adjust), patients <18 or >65 years old, pregnancy
- Adverse effects: Diarrhea, vomiting and abdominal pain
- Dosing (FYI): 2x333mg three times per day

- [ ] Both are safe to start in patients consuming alcohol, but more effective after withdrawal management

ALCOHOL USE DISORDER

2ND LINE AGENTS

2nd line Agents	Key Points
Disulfiram • Aldehyde dehydrogenase inhibitor	• Causes a "disulfiram reaction*" within 12-24 hours of consuming alcohol (ideally abstinent for 48 hours before initiating therapy) • Aversive therapy: fear of unpleasant reaction • LFTs monitored within first few weeks, then every 6 months • C/I: hepatic cirrhosis or insufficiency, ongoing alcohol use • Not commercially sold, may be available at select compounding pharmacies
Topiramate • Anticonvulsant	• Gradual titration over several weeks to avoid side effects • Lower doses and slower titration in CrCl <70 mL/min • SE: sedation, cognitive dysfunction, dizziness, paresthesia, and weight loss
Gabapentin • Anticonvulsant	• Lower doses recommended CrCl <60 mL/min • SE: somnolence, dizziness, ataxia, fatigue, diarrhea, nausea • Risk of misuse/addiction

*Disulfiram reaction: nausea, vomiting, flushing, headache, tachycardia

MONITORING PARAMETERS

EFFICACY ENDPOINTS				
Parameter	**Monitoring Tools**	**Frequency**	**Degree of Change**	**Timeframe**
Alcoholic drinks per week	Patient questionnaire BAC*	Every Visit	Decrease	Weeks

SAFETY ENDPOINTS				
Parameter	**Monitoring tool**	**Frequency**	**Degree of change**	**Timeframe**
Withdrawal seizures	CIWA-Ar PAWSS	Minimum daily	Prevent	First 48 hours of abstinence
Sedation	Patient questionnaire	At every appointment	Minimal to none	Duration of therapy
Cognitive dysfunction				
Nausea/vomiting				
Abdominal pain				

TAKEAWAY FOR OPIOID OVERDOSE AND AUD

Naloxone	• Opioid antagonist is given as an injection or nasal spray
	• Repeat 2nd dose in 2 to 3 minutes if first dose does not work
	• Must call 911 even if it appears to have worked
	  • Signs of overdose: slow or shallow breathing, blue nails or lips, constricted pupils, cold or clammy skin, unresponsiveness, drowsiness
	  • Remember to put the individual in recovery position following dose administration: head tilted back, hand supports the head, knee stops the body from rolling onto the stomach

First-line alcohol use disorder	• Acamprosate – recommended for patients with hepatic impairment and patient who aim for complete abstinence • Naltrexone – recommended for patients who aim for complete abstinence or reduced alcohol consumption

MYTHS/MISCONCEPTIONS

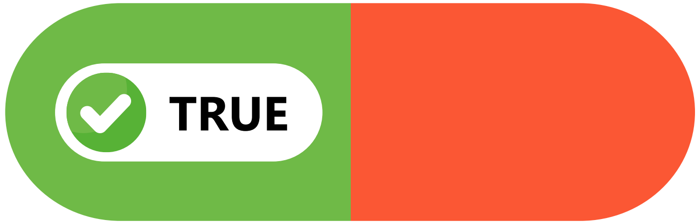
Naloxone will also reverse overdose caused by benzodiazepines, stimulants, and alcohol

Only long-term use of opioids is addictive

Naloxone is harmful when given to the “wrong person”

Opioids are effective in treating long-term chronic pain

MYTHS/MISCONCEPTIONS



Naloxone only blocks opioid actions
It will not reverse overdose involving alcohol, benzodiazepines, and stimulants

Short-term opioid use also has risks of addiction

Naloxone is safe, and is unlikely to create more harm if administered to someone who is unconscious due to non-opioid overdose

Long-term opioid use may lead to development of tolerance
Tolerance occurs when a higher dose of drug is needed to have an effect

CASE 1

NT is a 62-year-old retired lawyer with a recent diagnosis of opioid use disorder. She has just returned from her family doctor's office after discussing opioid agonist treatment options. She would like to read more about methadone and Suboxone® (buprenorphine/naloxone) and comes to your pharmacy to ask you about the risks and benefits of these medications. She has a medical history significant for hypertension, GERD, low back pain, and anxiety. She is currently taking HCTZ 25 mg daily, pantoprazole 20 mg daily, pregabalin 300 mg BID, clonazepam 1 mg every morning and 2 mg at bedtime, vitamin D 1000 IU daily, and a multivitamin daily. She does not smoke, but she has a history of alcohol dependence, she currently drinks 4 drinks per week. She has no allergies to medications. Today her blood pressure is 110/60 mmHg, her CrCl and LFTs are within normal limits, and she is reporting a pain intensity of 3/10.

1. Choose the most accurate statement:

   a) To minimize overdose risk, buprenorphine is in a fixed combination with naltrexone
   b) Suboxone® has a strong binding affinity to the kappa opioid receptor
   c) Suboxone® is considered to have more overdose risk compared to methadone
   d) Suboxone® may be preferred over methadone in patients with prolonged QT interval

CASE 1

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2. Which statement is false?

   a) Prescribers do not require an exemption to prescribe methadone
   b) Buprenorphine is available in an extended-release subcutaneous injection
   c) Methadone has higher risks for QTc prolongation than Suboxone®
   d) Methadone should be dispensed in 100 mL of water for opioid use disorder

CASE 1

NT is a 62-year-old retired lawyer with a recent diagnosis of opioid use disorder. She has just returned from her family doctor's office after discussing opioid agonist treatment options. She would like to read more about methadone and Suboxone® (buprenorphine/naloxone) and comes to your pharmacy to ask you about the risks and benefits of these medications. She has a medical history significant for hypertension, GERD, low back pain, and anxiety. She is currently taking HCTZ 25 mg daily, pantoprazole 20 mg daily, pregabalin 300 mg BID, clonazepam 1 mg every morning and 2 mg at bedtime, vitamin D 1000 IU daily and a multivitamin daily. She does not smoke, but she has a history of alcohol dependence, she currently drinks 4 drinks per week. She has no allergies to medications. Today her blood pressure is 110/60 mmHg, her CrCl and LFTs are within normal limits, and she is reporting a pain intensity of 3/10.

3. Which statement is true for opioid agonist treatment (OAT)?

   a) Abstinence based treatment is just as effective as OAT
   b) All patients are eligible for take-home doses after 12 weeks
   c) Pregnant women should be transitioned from Suboxone® to buprenorphine alone
   d) Methadone doses may be partially replaced if witnessed emesis occurs within 15 minutes of ingestion

CASE 1

NT is a 62-year-old retired lawyer with a recent diagnosis of opioid use disorder. She has just returned from her family doctor's office after discussing opioid agonist treatment options. She would like to read more about methadone and Suboxone® (buprenorphine/naloxone) and comes to your pharmacy to ask you about the risks and benefits of these medications. She has a medical history significant for hypertension, GERD, low back pain, and anxiety. She is currently taking HCTZ 25 mg daily, pantoprazole 20 mg daily, pregabalin 300 mg BID, clonazepam 1 mg every morning and 2 mg at bedtime, vitamin D 1000 IU daily and a multivitamin daily. She does not smoke, but she has a history of alcohol dependence, she currently drinks 4 drinks per week. She has no allergies to medications. Today her blood pressure is 110/60 mmHg, her CrCl and LFTs are within normal limits, and she is reporting a pain intensity of 3/10.

4. Choose the most accurate statement:

   a) Methadone may be more effective in severe or complex addictions
   b) Methadone is a mu receptor and NMDA agonist
   c) Methadone should be stopped during pregnancy as this drug crosses the placenta and can harm the baby
   d) Methadone can be rapidly titrated to the effective dose

CASE 2

DT is a 57-year-old male with a recent diagnosis of opioid use disorder. After speaking to you and reading about OAT he has decided to start Suboxone® (buprenorphine/naloxone). Two weeks later, he is stabilized on it and comes to your pharmacy for his regular observed dose. While there, he asks you about naloxone. He has a medical history significant for hypertension, hypercholesteremia, and opioid use disorder for which he takes ramipril 2.5 mg daily, atorvastatin 20 mg daily, and Suboxone® (buprenorphine/naloxone) 16/4 mg daily. He does not smoke; however, he has a history of alcohol dependence and currently drinks socially (2 drinks per week). He has no allergies to medications.

1. Which statement is the most accurate about naloxone?

   a) Naloxone blocks the respiratory depressive effects of morphine, alcohol, and lorazepam
   b) Naloxone is a strong opioid agonist
   c) If someone is not breathing, you can only use the injectable form
   d) Naloxone kits are available from most community pharmacies

CASE 2

DT is a 57-year-old male with a recent diagnosis of opioid use disorder. After speaking to you and reading about OAT he has decided to start Suboxone® (buprenorphine/naloxone). Two weeks later, he is stabilized on it and comes to your pharmacy for his regular observed dose. While there, he asks you about naloxone. He has a medical history significant for hypertension, hypercholesteremia, and opioid use disorder for which he takes ramipril 2.5 mg daily, atorvastatin 20 mg daily, and Suboxone® (buprenorphine/naloxone) 16/4 mg daily. He does not smoke; however, he has a history of alcohol dependence and currently drinks socially (2 drinks per week). He has no allergies to medications.

2. Which of the following is NOT a sign of an overdose?

   a) Constricted pupils
   b) Cold or clammy skin
   c) Alertness
   d) Slow, shallow breathing

CASE 2

DT is a 57-year-old male with a recent diagnosis of opioid use disorder. After speaking to you and reading about OAT he has decided to start Suboxone® (buprenorphine/naloxone). Two weeks later, he is stabilized on it and comes to your pharmacy for his regular observed dose. While there, he asks you about naloxone. He has a medical history significant for hypertension, hypercholesteremia, and opioid-use disorder for which he takes ramipril 2.5 mg daily, atorvastatin 20 mg daily, and Suboxone® (buprenorphine/naloxone) 16/4 mg daily. He does not smoke; however, he has a history of alcohol dependence and currently drinks socially (2 drinks per week). He has no allergies to medications.

3. Which of the following is NOT a harm reduction measure?

   a) Taking opioids with benzodiazepines
   b) Using drugs with others present rather than alone
   c) Naloxone kits
   d) Using needle exchange programs

CASE 2

DT is a 57-year-old male with a recent diagnosis of opioid use disorder. After speaking to you and reading about OAT he has decided to start Suboxone® (buprenorphine/naloxone). Two weeks later, he is stabilized on it and comes to your pharmacy for his regular observed dose. While there, he asks you about naloxone. He has a medical history significant for hypertension, hypercholesteremia, and opioid-use disorder for which he takes ramipril 2.5 mg daily, atorvastatin 20 mg daily, and Suboxone® (buprenorphine/naloxone) 16/4 mg daily. He does not smoke; however, he has a history of alcohol dependence and currently drinks socially (2 drinks per week). He has no allergies to medications.

4. What steps do you take during an opioid overdose?

   a) Stimulate, call 911, give naloxone, recovery position
   b) Stimulate, give naloxone, call 911, recovery position
   c) Stimulate, give CPR/chest compression, give naloxone, recovery position
   d) Give naloxone, stimulate, give CPR/chest compression, call 911

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CHANGE LOG

May 2025

- Added slides 10 and 17
- Slide 11: added "Preferred first line therapy" and edited graphic
- Slide 15: added "also preferred first-line therapy" per 2024 guideline updates
- Side 18: added first-line therapy
- Slide 19: added second-line therapy
- Slide 21: buprenorphine as first-line therapy, morphine SR as second-line therapy
- Slide 22: added "summary" to title
- Added slides 32-35

July 2025

- Formatting, grammar, and phrasing changes throughout
- Slide 7: Added harm reduction slide
- Slides 9 and 11: Moved information to fit more in theme with slides
- Removed probuphine slide as no longer available in Canada
- Removed opioid metabolism slide, as it is covered in pain management lecture
- Slide 30: Added to include harm reduction strategies for AUD treatment added specific phrasing is from the DSM-V
- Slide 34: Emphasized acamprosate can't be used if goal is just to decrease consumption of alcohol, marked dosing FYI
- Slide 35: Removed gabapentin dosing
- Slide 41: Added "partially" to the methadone replacement answer
- Slide 42: Changed "none of the above" to another answer option

November 2025

- Minor formatting changes made
- Slide 3: Updated legend
- Slide 15: Added information
- Slides 29-30 and 38: Expanded on monitoring parameters

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 33: Rephrased recommendations for clarity